Individual Plots:

1. Plot Type: Return Distribution (Histogram of Daily Returns)

- Across the set of top 20 high-volume stocks, the return distributions exhibit a consistent central tendency around zero, reflecting the absence of systematic daily drift and aligning with the efficient market expectation for short-term returns. However, beyond this common centre, there is substantial variation in dispersion and tail behaviour across stocks, indicating meaningful differences in risk profiles despite similar liquidity characteristics.
- A key pattern observed is the difference in the spread of the distributions. Some stocks display tightly concentrated distributions with a sharp peak around zero and narrow support, indicating low day-to-day variability and relatively stable price movements. In contrast, other stocks exhibit much wider distributions with flatter peaks, suggesting higher volatility and more frequent large price changes. This divergence highlights that high trading volume does not imply uniform stability, and that volatility is driven more by firm-specific and sector-specific dynamics than by liquidity alone.
- Another important feature is the presence of heavy tails across most distributions. Compared to a normal distribution, the empirical histograms show a higher frequency of extreme positive and negative returns. This indicates excess kurtosis and suggests that large market moves occur more often than would be predicted under a Gaussian assumption. The implication is that standard risk models that rely on normality may underestimate tail risk for these assets.
- There is also mild asymmetry in certain distributions, where either positive or negative returns extend further, indicating skewness. This reflects the possibility of asymmetric responses to market information, such as stronger reactions to negative news or speculative upward movements in growth-oriented stocks.
- Overall, the return distribution analysis reveals three key insights. First, daily returns are centred but not identical in variability across stocks. Second, there is clear evidence of non-normality through fat tails and peakedness. Third, even within a uniformly high-volume bucket, stocks exhibit heterogeneous risk characteristics, emphasising that volume alone is not a sufficient indicator of return behaviour or risk exposure.

2. Plot Type: Rolling Volatility (21-Day Window)

- Across the set of top 20 high-volume stocks, rolling volatility exhibits strong time variation rather than remaining constant, with clear clustering of high and low volatility periods. This indicates that market risk is dynamic and evolves, contradicting the assumption of constant variance often used in simpler financial models.
- A prominent pattern is the presence of volatility clustering. Periods of elevated volatility are not isolated spikes but occur in sustained intervals where large fluctuations persist for extended durations before gradually reverting to lower levels. Similarly, calm periods with low volatility tend to persist, creating distinct regimes of high and low risk. This persistence suggests that shocks to the market have lasting effects rather than dissipating immediately.
- Another important observation is the synchronisation of major volatility spikes across different stocks, particularly during known market stress periods. These spikes are significantly larger than baseline volatility levels and appear across multiple assets, indicating systemic events rather than firm-specific shocks. Despite differences in magnitude, the timing of these spikes is broadly aligned, implying that macroeconomic or market-wide factors dominate during crisis periods.

- At the same time, there is clear heterogeneity in baseline volatility levels across stocks. Some stocks consistently operate at higher volatility levels even during stable periods, while others remain relatively subdued. This suggests that while systemic events affect all stocks, the inherent risk profile of each stock determines how strongly it reacts and how quickly it stabilises afterwards.
- Additionally, the transition between high and low volatility regimes is not abrupt but gradual, with volatility building up before peaks and slowly decaying afterwards. This gradual adjustment reflects the accumulation and unwinding of market uncertainty over time rather than instantaneous changes.
- Overall, the rolling volatility analysis reveals three key insights. First, volatility is time-varying and exhibits strong clustering behaviour. Second, market-wide events drive synchronised spikes across stocks, highlighting systemic risk. Third, individual stocks maintain distinct baseline volatility levels, reinforcing that risk is influenced by both common market factors and asset-specific characteristics.

3. Plot Type: Daily Returns Time Series

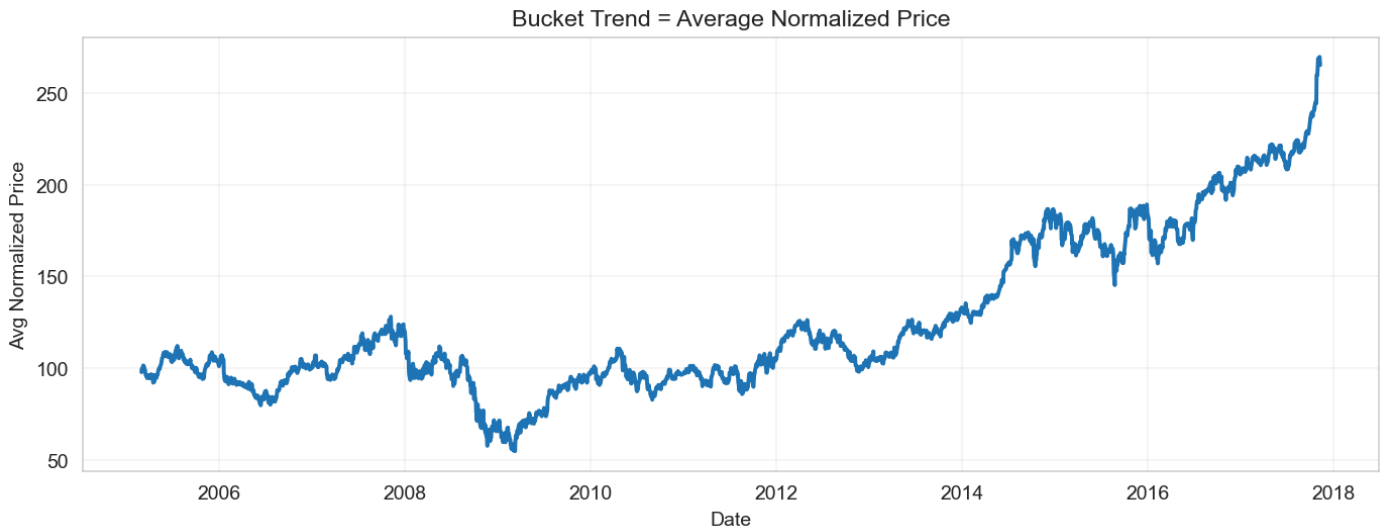
- Across the set of top 20 high-volume stocks, the daily returns time series shows that returns fluctuate randomly around zero with no visible long-term trend, which is consistent with the idea that short-term price movements are largely unpredictable. However, while the mean behaviour appears stable, the variability of returns changes significantly over time, revealing important underlying structure in market dynamics.
- A key pattern observed is the presence of volatility clustering. Periods with small, tightly packed returns are followed by intervals where large positive and negative spikes occur in rapid succession. These high-activity periods are not isolated but persist over stretches of time, indicating that large shocks tend to be followed by further large movements rather than immediate stabilisation. This aligns with the earlier observation from rolling volatility that risk is time-dependent and persistent.
- Another important feature is the occurrence of extreme spikes in returns. These spikes represent sudden large gains or losses and are noticeably more frequent during certain time windows. These events are not evenly distributed across the timeline but concentrated in specific periods, suggesting the influence of market-wide shocks or major economic events. The magnitude of these spikes varies across stocks, with some exhibiting significantly larger jumps, indicating differences in sensitivity to new information.
- There is also clear heterogeneity in how different stocks respond during turbulent periods. While all stocks show increased activity during high-volatility phases, some display much larger swings and more frequent extreme movements, whereas others remain relatively contained. This indicates that even under similar market conditions, individual stocks exhibit different levels of responsiveness and risk exposure.
- Additionally, the transition between calm and turbulent periods is gradual rather than instantaneous. Periods of increasing activity build up before reaching peak instability and then slowly revert to lower variability. This suggests that market uncertainty accumulates over time and dissipates progressively rather than abruptly.
- Overall, the daily returns time series highlights three key insights. First, returns are centred around zero with no predictable trend. Second, volatility clustering is clearly present, with periods of high and low activity persisting over time. Third, extreme return events occur intermittently and are concentrated in specific periods, reinforcing the presence of time-varying risk and heterogeneous behaviour across stocks.

4. Plot Type: Normalised Price Over Time

- Across the set of top 20 high-volume stocks, normalised price trajectories reveal strong long-term upward trends, but with substantial variation in growth patterns, drawdowns, and recovery dynamics. While most stocks exhibit overall appreciation over extended horizons, the paths taken to reach these levels differ significantly, highlighting differences in growth potential, resilience, and exposure to market cycles.
- A key pattern is the presence of sustained long-term growth in several stocks, where prices increase by large multiples over time. These trajectories are not linear but exhibit periods of accelerated growth followed by consolidation or correction phases. This indicates that returns are driven by compounding over time, but growth occurs in bursts rather than at a constant rate. Some stocks demonstrate exponential-like growth patterns, suggesting strong market confidence and persistent positive momentum.
- At the same time, there are clear instances of sharp drawdowns, where prices fall significantly from their peaks before stabilising or recovering. These declines are often abrupt and substantial, indicating vulnerability to market-wide downturns or sector-specific shocks. The magnitude of these drawdowns varies across stocks, with some experiencing deep collapses followed by slow recoveries, while others show quicker rebounds, reflecting differences in underlying strength and investor sentiment.
- Another important feature is the variation in recovery behaviour following downturns. Some stocks recover to previous highs and continue to grow, indicating resilience and long-term value creation. Others fail to regain earlier peaks within the observed period, suggesting structural challenges or weaker long-term performance. This divergence highlights that not all high-volume stocks deliver consistent long-term returns, even if they are actively traded.
- Additionally, price paths show periods of relative stagnation or sideways movement, where growth slows, and prices fluctuate within a range. These phases often follow periods of rapid expansion or precede new growth cycles, indicating transitions between different market regimes.
- Overall, the normalised price analysis reveals three key insights. First, most high-volume stocks exhibit strong long-term growth driven by compounding, but with non-linear trajectories. Second, significant drawdowns and varying recovery patterns indicate differing levels of resilience across stocks. Third, despite belonging to the same high-volume category, stocks display heterogeneous long-term performance, emphasising that liquidity does not guarantee consistent growth outcomes.

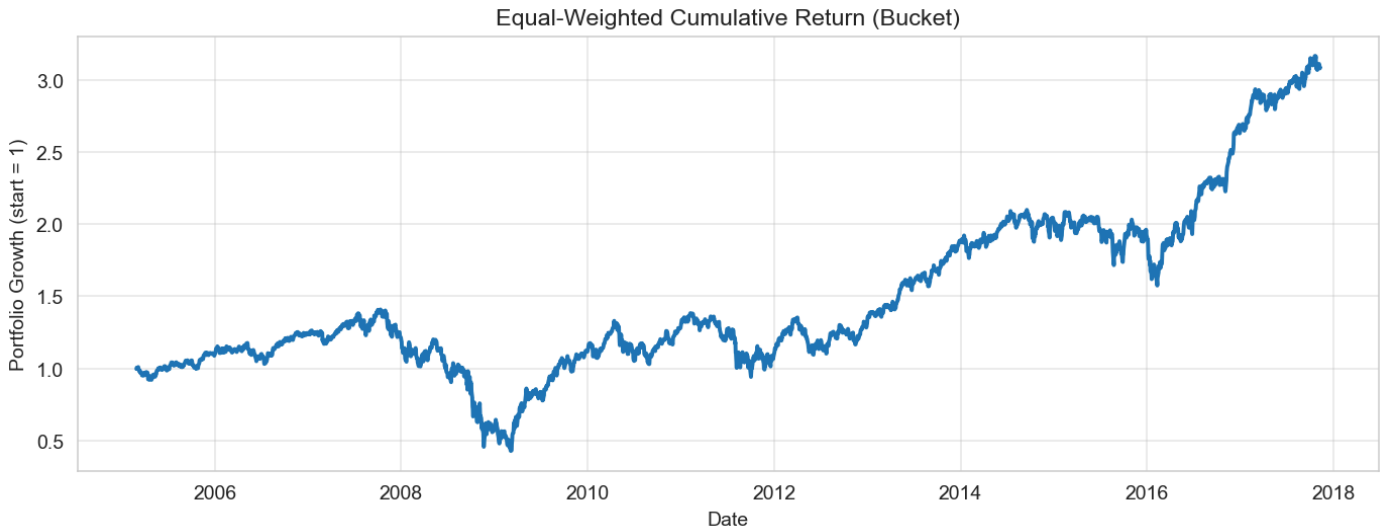
Bucket-Level Plots:

1. Plot Type: Average Normalized Price



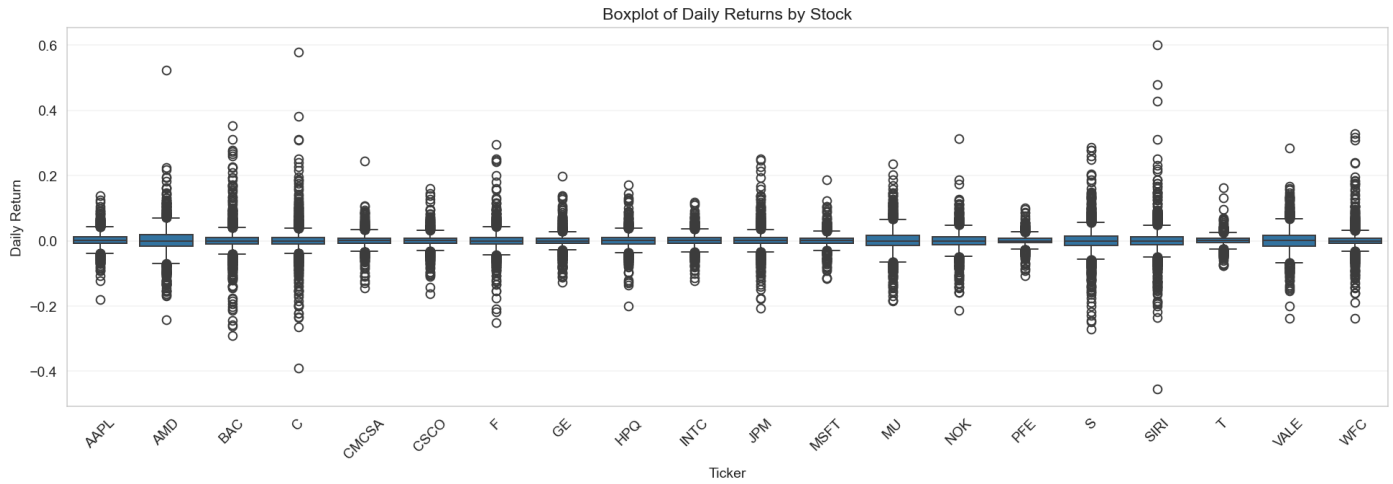
The average normalized price plot shows the overall movement of the stock bucket after putting all stocks on the same starting scale. The bucket dropped sharply during the 2008-2009 financial crisis, but it gradually recovered and continued to rise strongly in later years. This means that although the stocks faced serious short-term market pressure, the group as a whole still showed strong long-term growth. Overall, the plot suggests that the bucket was affected by major market downturns but remained upward-trending over time.

2. Plot Type: Equal-Weighted Cumulative Return



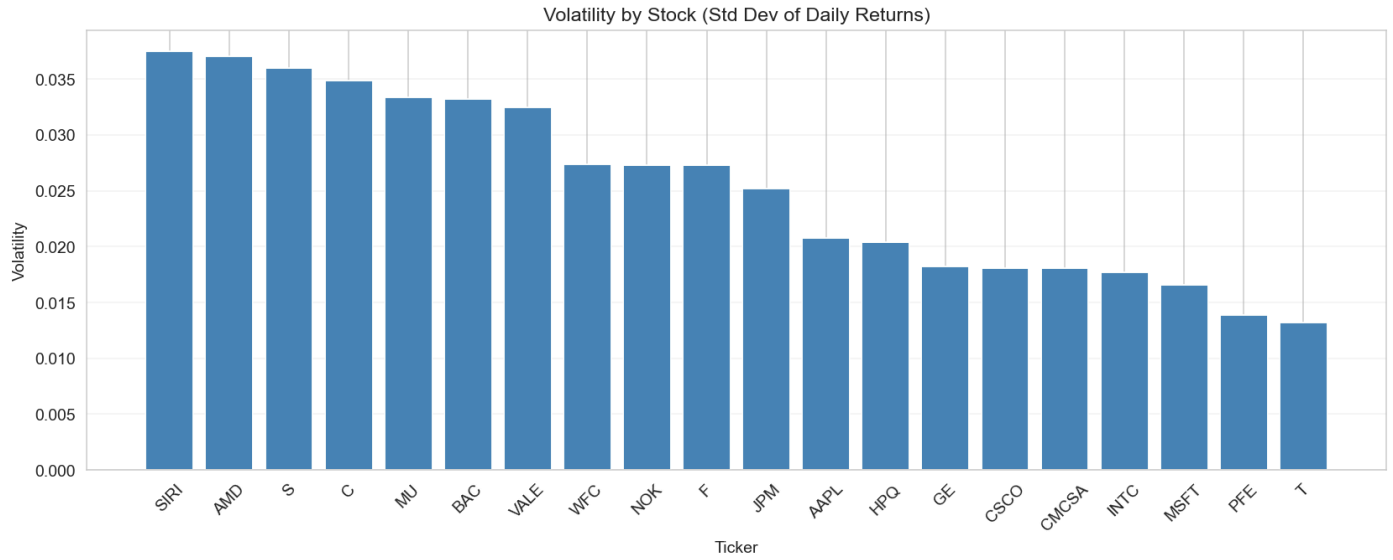
The equal-weighted cumulative return plot shows how a portfolio would perform if the same amount were invested in each stock. The portfolio suffered a large loss during the crisis period, but it recovered steadily and ended much higher than where it started. This suggests that the bucket had good long-term return potential, even though it experienced major short-term losses along the way. In other words, the portfolio rewarded long-term holding, but not without risk.

3. Plot Type: Boxplot of Daily Returns



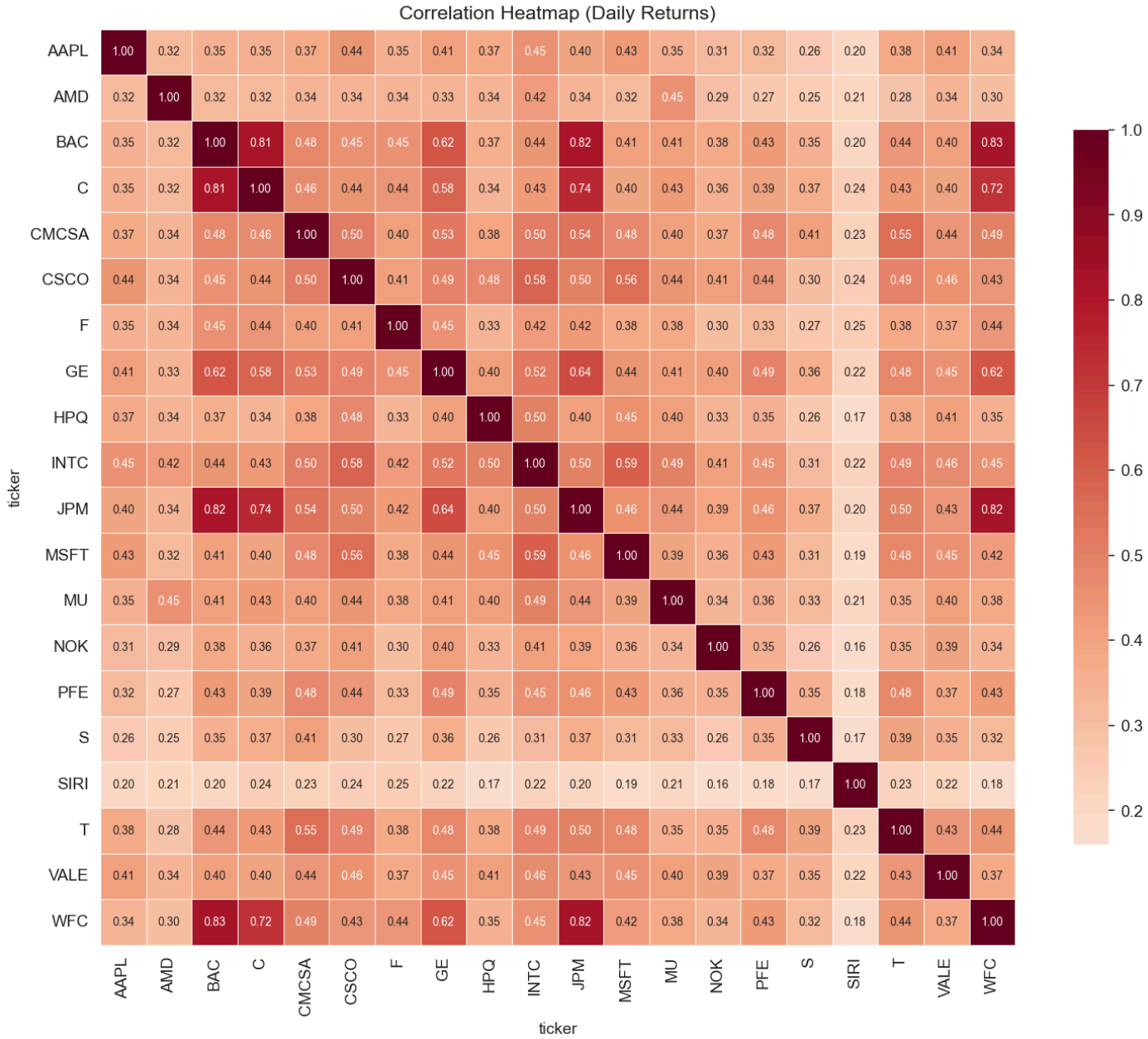
The boxplot shows that daily returns for most stocks are centered around zero, which is typical for high-frequency stock return data. However, several stocks have much wider spreads and many extreme outliers, meaning their returns are more dispersed and prone to unusually large moves. In particular, BAC, C, F, S, and SIRI exhibit stronger tail behavior, indicating greater exposure to sudden positive and negative price jumps. Overall, this suggests that return distributions are not well described by a normal distribution; instead, they display fat tails and occasional shock events.

4. Plot Type: Volatility Bar Chart



The volatility chart compares how much each stock's daily returns fluctuate. Some stocks, such as SIRI, AMD, and S, are much more volatile, while others like T, PFE, and MSFT are relatively stable. This shows that the bucket includes both high-risk and lower-risk stocks. As a result, not all stocks in the bucket behave the same way, and some may be much harder to predict than others.

5. Plot Type: Correlation Heat Map



The correlation heatmap shows how similarly the stocks move from day to day. Most stocks have positive correlations, meaning they tend to move in the same general direction, at least to some extent. Financial stocks such as BAC, C, JPM, and WFC are especially strongly related, which suggests that they are influenced by similar sector-level factors. On the other hand, SIRI is less correlated with the others, meaning its movements are more independent. This shows that the bucket has some diversification, but many stocks are still affected by common market forces.